



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B.Sc Chemistry				
Course Title	COMMERCIAL POLYMERS				
Type of Course	VOCATIONAL MINOR				
Semester	II				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3		2	75
Pre-requisites	Concept of polymers. Use of polymers. Knowledge on synthetic polymers and their application.				
Course Summary	This course is intended to provide basic knowledge about commercial polymers. It deals with types, techniques of preparation and characterization of plastics, rubber and fibre materials. The applications of these materials in daily life, engineering and biomedical fields have been emphasized. The students are exposed to the problems of polymer waste management and the strategies developed to minimize plastic pollution.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic principles of polymer chemistry, including polymerization mechanisms, molecular structure, and polymer characterization techniques.	U	C	Instructor-created exams / Quiz
CO2	Students should become familiar with the major families of commercial polymers, such as polyethylene, polypropylene, polyvinyl chloride (PVC), polystyrene, polyethylene terephthalate (PET), and others.	R,U	F	Class test /Assignment / Quiz
CO3	Students should gain awareness of the environmental impact of polymers and explore sustainable practices in polymer production, usage, and disposal. They should understand concepts such as biodegradability, recycling, and the development of eco-friendly polymers.	R, U, Ap	F	Class test /Assignment / Quiz

CO4	Students should be able to correlate the molecular structure of thermoplastics with their physical, mechanical, thermal, electrical, and chemical properties. They should understand how these properties influence the selection of thermoplastics for specific applications.	R, U, Ap	C	Class test /Assignment / Quiz
CO5	Students should learn about the synthesis methods and curing mechanisms used for thermosetting polymers, including techniques such as condensation polymerization, addition polymerization, and crosslinking reactions.	R, U	P	Class test /Assignment / Quiz
CO6	Students should be able to classify elastomers based on their chemical structure, polymerization mechanisms, and properties. They should also learn methods for identifying different types of elastomers	R, U	P	Class test /Assignment / Quiz
CO7	Develop practical skill in the preparation of various types of polymers	Ap	P	Viva/ Practical Skill
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hours	Mark
I	INTRODUCTION TO COMMERCIAL POLYMERS		10	22
	1	Types of polymer (Thermoplastics, Thermosetting polymers, Elastomers, Fibers).	1	
	2	Production Methods, Polymerization, Extrusion, Injection Molding, Blow Molding.	4	
	3	Properties and Applications, Mechanical Properties, Thermal Properties, Chemical Resistance, Electrical Properties, Applications.	3	
	4	Environmental Considerations.	2	
II	THERMOPLASTICS AND THERMOSETS		17 Hrs	36
	5	Polyolefins and allied polymers, Vinyl polymers, Styrene and its copolymers.	3	

	6	Acrylics, Polyamides, Polyesters, PU, Fluoropolymers.	3	
	7	Cellulose and its derivatives, Polycarbonates, Polyacetals.	2	
	8	PES, PEI, PEEK, Polyacrylic acid, PVA, Polyvinyl Acetals.	2	
	9	Thermosetting plastics, Phenol Formaldehyde, Melamine Formaldehyde.	2	
	10	Urea Formaldehyde, Epoxy resins, Unsaturated polyester.	2	
	11	Vinyl esters, Cyanate esters.	1	
	12	Furan resins and Silicone polymers.	2	
III	ELASTOMERS		8 Hrs	18
	13	Lignins, Cellulose and its derivatives, Chitin, Chitosan, Source, Properties and Applications.	2	
	14	Reclaimed rubber, Reclaiming processes.	1	
	15	Elastomers, Natural Rubber, Isoprene rubber, Modified forms of NR- butyl rubber, Nitrile rubber, Chloroprene Rubber.	2	
	16	Styrene-Butadiene Rubber, EPDM.	1	
	17	Vulcanization, Rubber chemicals, Thermoplastic elastomers.	2	
IV	FIBERS		10 Hrs	22
	18	Classification, Sources of fibers, Essential properties of textile fibers.	1	
	19	Sources of cellulose, Sources of cellulosic fibers, Sources of synthetic fibers.	2	
	20	Fibers formation, Synthesis of monomer, Polymerization and Formation of polymer, Characteristics of fibers formation polymers, Drawing.	2	
	21	Fibers structure: Unit cell, Arrangement of chain molecules in the crystallites, Formation and arrangement of crystallites in fibers, Chemical methods.	3	
	22	Vegetable fibers, Cellulose, Jute, Flax, Hemp, Ramie, Sisal, Pineapple, Coir.	2	
V	PRACTICAL		30	
		Any Five preparations of the following. 1. Preparation of Glyptal resin 2. Preparation of phenol formaldehyde		

		resin 3. Preparation of urea formaldehyde resin 4. Preparation of aniline formaldehyde resin 5. Preparation of polyaniline 6. Preparation of Nylon 6,6 7. Preparation of PMMA 8. Preparation of linear polystyrene (free radical polymerization) 9. Preparation of crosslinked polystyrene (suspension polymerization) Teacher can suggest preparation of any polymers other than mentioned above (Open ended)		
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References

1. "Introduction to Polymers" by Robert J. Young and Peter A. Lovell
2. "Handbook of Thermoplastics" edited by Olagoke Olabisi and Kolapo Adewale.
3. "Thermoplastic Materials: Properties, Manufacturing Methods, and Applications" by Christopher C. Ibeh
4. "Thermosetting Polymers: Synthesis, Properties, and Applications" edited by Ulf W. Gedde
5. "Thermosets: Structure, Properties, and Applications" edited by Qipeng Guo.
6. "Introduction to Thermosetting Plastics" by Syed Qutubuddin and Aftab Ahmed
7. "Elastomers: Types, Properties and Applications" edited by Aubrey Q. Stokes
8. "Rubber Technology Handbook" by Werner Hofmann and Walter Holzwarth
9. "Elastomer Technology Handbook" edited by Jiri George Drobny
10. "Textile Fibre Structure" by M.A. Hearle, W.E. Morton, and B.S. Harwood

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	-	-	1	1	1	2	-	1	-	1	-	1
CO 2	2	-	-	1	2	1	2	-	1	-	1	1	1
CO 3	3	-	1	3	1	3	2	-	2	-	2	3	2
CO 4	2	-	1	1	1	2	2	-	2	1	2	2	1

CO 5	1	-	-	1	1	1	1	-	1	-	1	-	1
CO 6	2	-	-	2	1	2	2	-	-	-	1	1	1
CO 7	2	-	2	1	1	1	2	-	1	-	-	-	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)
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Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/ Seminar/ Viva	Practical skill Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓

CO 6	✓	✓		✓
CO 7		✓	✓	